

Marcel Wagenländer

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Profile

ML systems researcher and engineer focused on scaling large language models. PhD candidate at Imperial College London under Peter Pietzuch (LSDS), first author of **Tenplex** (SOSP'24) on dynamic parallelism for distributed ML training, and of **Scepsy** (under submission) on serving agentic LLM workflows. Research Scientist intern at Meta in summer 2025, modelling block-diffusion language model serving inside Meta's internal performance simulator. Working across the stack, from Triton kernels through PyTorch to cluster orchestration. Looking to build the training and inference systems behind frontier LLMs.

Research Experience

Meta Platforms, Inc. **Menlo Park, United States**
Research Scientist Intern Jun 2025 - Sep 2025

- Showed that **block-diffusion language models** incur no extra latency from their additional compute, because attention is memory-bound
- Extended Meta's internal performance simulator to model block-diffusion serving, enabling production teams to evaluate throughput and latency trade-offs against autoregressive baselines

Imperial College London, Large-scale Data & Systems Group (LSDS) **London, United Kingdom**
Research Assistant Oct 2019 - Mar 2020

- First-author HotCloud'20 workshop paper (**Spotnik**, 30+ citations) on adaptive synchronisation for distributed deep learning on transient cloud resources
- Co-authored **KungFu** (OSDI'20, 110+ citations), an open-source distributed ML framework now with 295+ GitHub stars

Education

Imperial College London **London, United Kingdom**
PhD candidate in Computing Jul 2021 - Expected Jun 2026

- Supervised by Prof. Peter Pietzuch in the Large-scale Data & Systems (LSDS) group
- **Tenplex** (SOSP'24, first author, 40+ citations): dynamic parallelism for distributed deep-learning training, evaluated on GPT-3 models up to 6.7B parameters across 16- to 32-GPU clusters. 18% reduction in time-to-target-step vs. PyTorch and Tenplex-DP baselines
- **Scepsy** (first author, under submission): efficient GPU scheduling for agentic LLM workflows

Technical University of Munich **Munich, Germany**
Master of Science in Informatics Apr 2018 - Jun 2021

- Thesis: **Memory Management for Graph Neural Networks on GPUs**. Offloading techniques for training GNNs on graphs that exceed GPU memory
- Six-person team project: implemented autonomous parking using ROS, LIDAR, and SLAM

Technical University of Munich **Munich, Germany**
Bachelor of Science in Informatics Oct 2014 - Mar 2018

- Thesis: **Classification of Goalkeeper Movements**. Predicted exercises from inertial measurement unit data with 80% accuracy
- Nine-person team project for T-Systems: developed a smart-textile system for in-field data gathering

Deakin University **Melbourne, Australia**
Study Abroad (BSc exchange) in Computer Science Feb 2017 - Jun 2017

- Programmed a Nao humanoid robot to execute autonomous parcel delivery

Publications

- [1] M. Wagenländer* et al., "Scepsy: Serving Agentic Workflows Using Aggregate LLM Pipelines." [Online]. Available: <https://arxiv.org/abs/2604.15186>
- [2] M. Wagenländer, G. Li, B. Zhao, L. Mai, and P. Pietzuch, "Tenplex: Dynamic Parallelism for Deep Learning using Parallelizable Tensor Collections," in *Proceedings of the ACM SIGOPS 30th Symposium on Operating*

Systems Principles, in SOSP '24. Austin, TX, USA: Association for Computing Machinery, 2024, pp. 195–210. doi: 10.1145/3694715.3695975.

- [3] S. W. Ober*, A. Artemev*, M. Wagenländer*, R. Grobins, and M. van der Wilk, "Recommendations for Baselines and Benchmarking Approximate Gaussian Processes." [Online]. Available: <https://arxiv.org/abs/2402.09849>
- [4] L. Mai, G. Li, M. Wagenländer, K. Fertakis, A.-O. Brabete, and P. Pietzuch, "KungFu: Making Training in Distributed Machine Learning Adaptive," in *14th USENIX Symposium on Operating Systems Design and Implementation (OSDI 20)*, USENIX Association, Nov. 2020, pp. 937–954. [Online]. Available: <https://www.usenix.org/conference/osdi20/presentation/mai>
- [5] M. Wagenländer, L. Mai, G. Li, and P. Pietzuch, "Spotnik: Designing Distributed Machine Learning for Transient Cloud Resources," in *12th USENIX Workshop on Hot Topics in Cloud Computing (HotCloud 20)*, USENIX Association, July 2020. [Online]. Available: <https://www.usenix.org/conference/hotcloud20/presentation/wagenl%C3%A4nder>

Open-source Projects

Tenplex

github.com/kungfu-team/tenplex

Go state-management library (~6,700 LoC) for changing the parallelism strategy (tensor, data, pipeline) of distributed deep-learning jobs at runtime, with low reconfiguration overhead. Integrates with PyTorch, Megatron-LM, and DeepSpeed.

KungFu

github.com/llds/kungfu

Go/C++/Python adaptive distributed ML framework for TensorFlow, PyTorch, and MindSpore, with minimal code changes and no parameter servers. Co-author of the OSDI'20 paper; 295+ GitHub stars.

Selected Talks

- 30th ACM Symposium on Operating Systems Principles (SOSP'24) Austin, TX, USA, Nov 2024
Conference talk: *Tenplex: Dynamic Parallelism for Deep Learning using Parallelizable Tensor Collections*
- 12th USENIX Workshop on Hot Topics in Cloud Computing (HotCloud'20) Virtual, Jul 2020
Workshop talk: *Spotnik: Designing Distributed Machine Learning for Transient Cloud Resources*

Awards

- SOSP'23 Student Travel Grant (ACM SIGOPS) Koblenz, Germany, Oct 2023

Supervision

- Co-supervised Dhyey Trivedi (MEng, 2026): *KAIROS: Workflow-Aware GPU Autoscaling for Multi-LLM Pipelines*
- Co-supervised Jiaju Yang (MEng, 2025): *Load-Aware Optimisation of Retrieval-Augmented Generation Systems for Precision-Critical Applications*
- Co-supervised Tom Underhill (BEng, 2025): *Systems Optimisation for Inference-Time Scaling LLMs*
- Co-supervised Hamish McCreanor (MEng, 2025): *A New Scalable Runtime for LLM Inference*

Skills

- **Languages:** German (Native), English (Fluent), Spanish (Basic), Chinese (Basic)
- **ML Systems:** Distributed training with multi-dimensional parallelism, Dynamic parallelism and cluster reconfiguration, LLM and workflow serving, GPU performance analysis, Scheduling for ML workloads
- **ML Frameworks:** PyTorch, JAX, Megatron-LM, DeepSpeed, vLLM, SGLang, NumPy, Pandas
- **GPU Programming:** CUDA, Triton
- **Programming Languages:** Python, Go, Bash, Rust, C/C++, Swift
- **Infrastructure:** On-premise clusters, Cloud, Docker, Kubernetes
- **Tools:** Git, LaTeX, Matplotlib, OmniGraffle
- **Research Interests:** Distributed systems for ML, LLM training and inference infrastructure, ML model scaling, Dynamic resource allocation, Scheduling
- **Interests:** Running, Reading, Piano, Travelling, Skiing, Sailing